

utfcode
utf8.sty 3.10 UTF-8 input encoding 13.06.2000
scanner for code UTF-8 installed.

OmniTech Retail Solutions

SmartHome Hub v2 Network Architecture & Setup Manual

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1

1. Network Prerequisites and Protocol Support

The SmartHome Hub v2 acts as a localized IoT bridge designed to unify cross-brand home automation hardware.

1.1

1.1 Wireless Frequency and Pairing Rules

- **2.4GHz Wi-Fi Requirement:** Initial device pairing **requires a 2.4GHz Wi-Fi network connection**. 5GHz-only SSIDs are strictly unsupported during the initial setup phase.
- **Ethernet Fallback:** Includes a Gigabit RJ45 port for hardwired connections; highly recommended for setups exceeding 30 connected node devices.
- **Integrated Radios:** Thread, Matter 1.2, Zigbee 3.0, and Z-Wave Plus (868.4MHz / 908.4MHz band adaptive).

Table 9.1: Protocol Support and Node Limits

Protocol	Max Concurrent Nodes	Line-of-Sight Range	Mesh Support
Matter over Thread	128 Nodes	30 Meters	Native Mesh Routing
Zigbee 3.0	64 Nodes	20 Meters	Requires Powered Routers
Z-Wave Plus	232 Nodes	40 Meters	Native Mesh Routing

2 2. Local Processing vs. Cloud Synchronization

2.1 2.1 Edge Computing and Local Automation Scripts

To guarantee operational reliability during internet service provider (ISP) outages:

- **Local Execution:** All user-defined automation triggers, lighting routines, and sensor rules execute locally on the internal eMMC edge storage without relying on external cloud endpoints.
- **Cloud Sync Schedule:** Encrypted system configurations and security snapshots are backed up to OmniTech Secure Cloud every 24 hours at **03:00 AM UTC**.

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